

TE Curve Verification Pipeline Selected Active Model Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates only the currently selected active model families against the repository held-out TE-curve test split. The `global` surface is intentionally excluded from this reduced decision report.

Dataset And Split

- dataset config: `config/datasets/transmission_error_dataset.yaml` ;
- dataset root: `data\polished_dataset` ;
- comparison mode: `selected_active_model_matrix` ;
- candidate count: `7` ;
- held-out curve count before candidate filtering: `100` ;
- percentage-error denominator: `peak_to_peak_truth` ;
- evaluated surface: `Fw` ;
- evaluated direction: `forward` ;

Candidate Inventory

Candidate	Source	Family	Surface	Valid Directions
<code>polished_actual_values_feedforward_Fw</code>	<code>polished_actual_values_model_archive</code>	<code>feedforward</code>	<code>Fw</code>	<code>forward</code>
<code>polished_actual_values_tree_Fw</code>	<code>polished_actual_values_model_archive</code>	<code>tree</code>	<code>Fw</code>	<code>forward</code>
<code>polished_actual_values_harmonic_regression_Fw</code>	<code>polished_actual_values_model_archive</code>	<code>harmonic_regression</code>	<code>Fw</code>	<code>forward</code>
<code>polished_actual_values_periodic_mlp_harmonic_Fw</code>	<code>polished_actual_values_model_archive</code>	<code>periodic_mlp_harmonic</code>	<code>Fw</code>	<code>forward</code>
<code>polished_actual_values_periodic_gru_sequence_Fw</code>	<code>polished_actual_values_model_archive</code>	<code>periodic_gru_sequence</code>	<code>Fw</code>	<code>forward</code>
<code>polished_actual_values_wave4_1_mae_robust_loss_Fw</code>	<code>polished_actual_values_model_archive</code>	<code>wave4_1_mae_robust_loss</code>	<code>Fw</code>	<code>forward</code>
<code>polished_actual_values_wave4_2_quantile_p10_p50_p90_Fw</code>	<code>polished_actual_values_model_archive</code>	<code>wave4_2_quantile_p10_p50_p90</code>	<code>Fw</code>	<code>forward</code>

Exact Model Paths

Candidate	Family	Surface	ONNX Model Path	Python Model Path
polished_actual_values_feedforward_Fw	feedforward	Fw	models/polished_dataset/actual_values/exported/feedforward/forward/2026-07-07-18-43-03_te_feedforward_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/feedforward/forward/2026-07-07-18-43-03_te_feedforward_fw_polished_actual_values/python/feedforward-epoch=181-val_mae=0.00161552.ckpt
polished_actual_values_tree_Fw	tree	Fw	models/polished_dataset/actual_values/exported/tree/forward/2026-07-07-09-55-31_te_tree_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/tree/forward/2026-07-07-09-55-31_te_tree_fw_polished_actual_values/python/tree_model.pkl
polished_actual_values_harmonic_regression_Fw	harmonic_regression	Fw	models/polished_dataset/actual_values/exported/harmonic_regression/forward/2026-07-08-00-15-14_te_harmonic_regression_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/harmonic_regression/forward/2026-07-08-00-15-14_te_harmonic_regression_fw_polished_actual_values/python/harmonic_regression-epoch=073-val_mae=0.00182314.ckpt
polished_actual_values_periodic_mlp_harmonic_Fw	periodic_mlp_harmonic	Fw	models/polished_dataset/actual_values/exported/periodic_mlp_harmonic/forward/2026-07-08-03-17-38_te_periodic_mlp_harmonic_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/periodic_mlp_harmonic/forward/2026-07-08-03-17-38_te_periodic_mlp_harmonic_fw_polished_actual_values/python/periodic_mlp-epoch=045-val_mae=0.00131065.ckpt
polished_actual_values_periodic_gru_sequence_Fw	periodic_gru_sequence	Fw	models/polished_dataset/actual_values/exported/periodic_gru_sequence/forward/2026-07-09-00-29-12_te_periodic_gru_sequence_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/periodic_gru_sequence/forward/2026-07-09-00-29-12_te_periodic_gru_sequence_fw_polished_actual_values/python/periodic_gru_sequence-epoch=200-val_mae=0.00150079.ckpt
polished_actual_values_wave4_1_mae_robust_loss_Fw	wave4_1_mae_robust_loss	Fw	models/polished_dataset/actual_values/exported/wave4_1_mae_robust_loss/forward/2026-07-12-18-19-09_te_wave4_1_mae_robust_loss_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_1_mae_robust_loss/forward/2026-07-12-18-19-09_te_wave4_1_mae_robust_loss_fw_polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=202-val_mae=0.00173420.ckpt
polished_actual_values_wave4_2_quantile_p10_p50_p90_Fw	wave4_2_quantile_p10_p50_p90	Fw	models/polished_dataset/actual_values/exported/wave4_2_quantile_p10_p50_p90/forward/2026-07-14-16-15-03_te_wave4_2_quantile_p10_p50_p90_fw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_2_quantile_p10_p50_p90/forward/2026-07-14-16-15-03_te_wave4_2_quantile_p10_p50_p90_fw_polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=250-val_mae=0.00176755.ckpt

Metric Ranking

Rank	Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
1	polished_actual_values_wave4_1_mae_robust_loss_Fw	0.001681	0.002032	3.355	8.132
2	polished_actual_values_periodic_gru_sequence_Fw	0.001676	0.002006	3.366	7.871
3	polished_actual_values_wave4_2_quantile_p10_p50_p90_Fw	0.001719	0.002071	3.453	8.191
4	polished_actual_values_periodic_mlp_harmonic_Fw	0.001900	0.002239	3.891	10.013
5	polished_actual_values_tree_Fw	0.002120	0.002599	4.341	8.662
6	polished_actual_values_feedforward_Fw	0.002181	0.002646	4.491	8.951
7	polished_actual_values_harmonic_regression_Fw	0.002355	0.002816	4.909	10.791

Direction Breakdown

Direction	Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
forward	polished_actual_values_wave4_1_mae_robust_loss_Fw	0.001681	0.002032	3.355	8.132
forward	polished_actual_values_periodic_gru_sequence_Fw	0.001676	0.002006	3.366	7.871
forward	polished_actual_values_wave4_2_quantile_p10_p50_p90_Fw	0.001719	0.002071	3.453	8.191
forward	polished_actual_values_periodic_mlp_harmonic_Fw	0.001900	0.002239	3.891	10.013
forward	polished_actual_values_tree_Fw	0.002120	0.002599	4.341	8.662
forward	polished_actual_values_feedforward_Fw	0.002181	0.002646	4.491	8.951
forward	polished_actual_values_harmonic_regression_Fw	0.002355	0.002816	4.909	10.791

Curve Evidence

Each selected candidate is shown with the same four deterministic held-out operating conditions for this direction. The dark line is the measured TE curve and the blue line is the model prediction for the same operating condition.

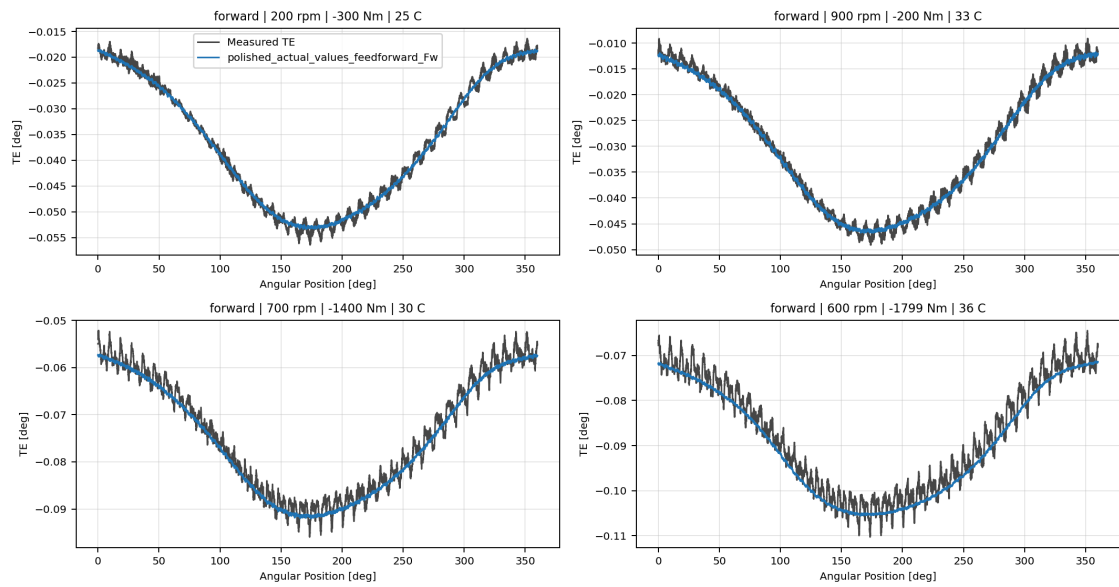
Shared operating conditions

- 200 rpm, 300 Nm, 25 C
- 900 rpm, 200 Nm, 30 C
- 700 rpm, 1400 Nm, 30 C
- 600 rpm, 1800 Nm, 35 C

Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
polished_actual_values_feedforward_Fw	Selected candidate	0.002181	0.002646	4.491	8.951
polished_actual_values_tree_Fw	Selected candidate	0.002120	0.002599	4.341	8.662
polished_actual_values_harmonic_regression_Fw	Selected candidate	0.002355	0.002816	4.909	10.791
polished_actual_values_periodic_mlp_harmonic_Fw	Selected candidate	0.001900	0.002239	3.891	10.013
polished_actual_values_periodic_gru_sequence_Fw	Selected candidate	0.001676	0.002006	3.366	7.871
polished_actual_values_wave4_1_mae_robust_loss_Fw	Surface leader	0.001681	0.002032	3.355	8.132
polished_actual_values_wave4_2_quantile_p10_p50_p90_Fw	Selected candidate	0.001719	0.002071	3.453	8.191

polished_actual_values_feedforward_Fw

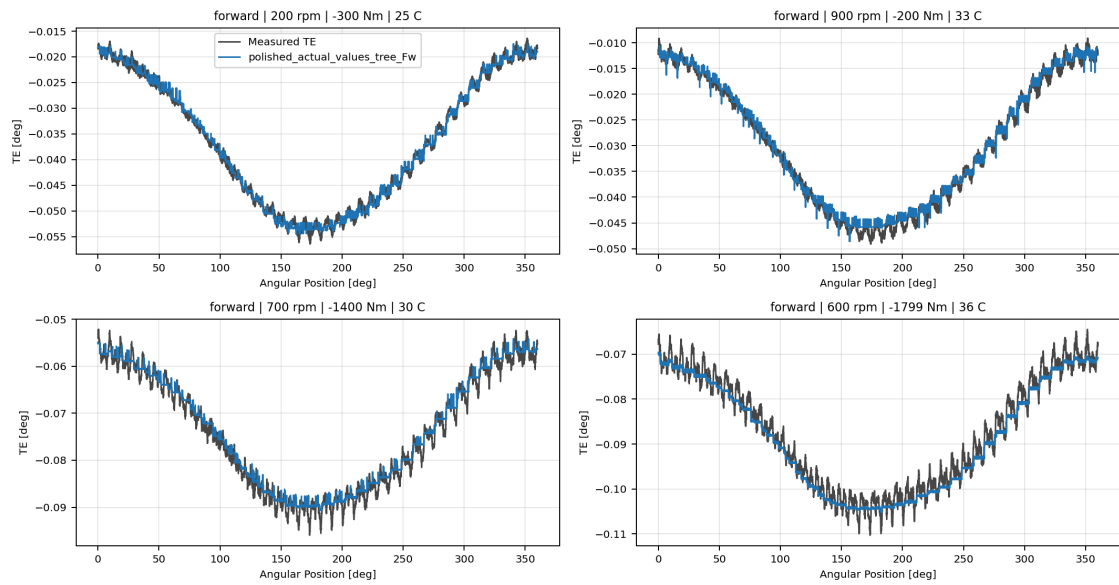
polished_actual_values_feedforward_Fw



polished_actual_values_feedforward_Fw measured-versus-predicted TE collage

polished_actual_values_tree_Fw

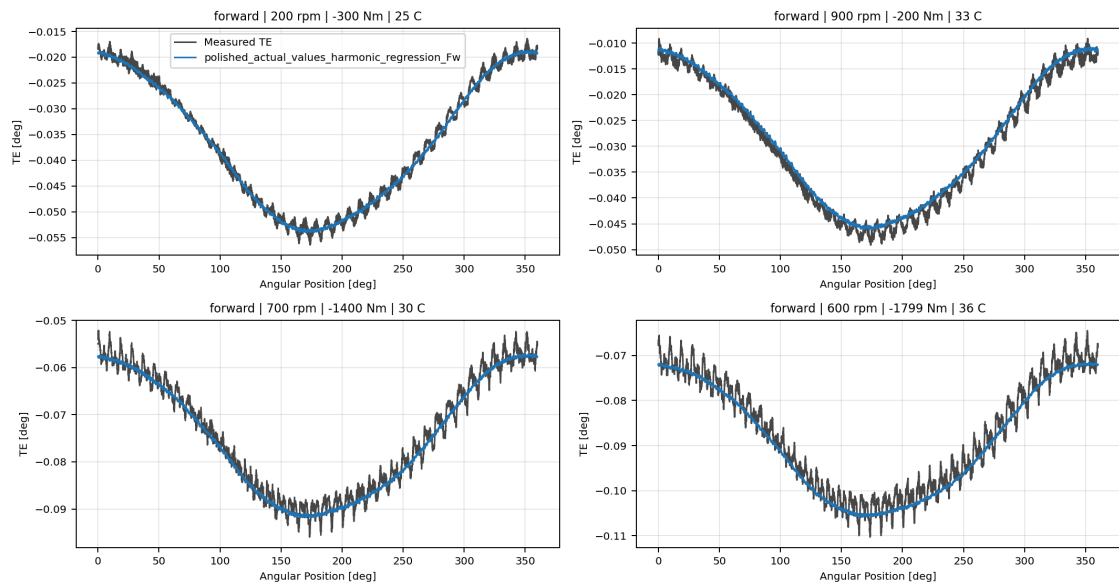
polished_actual_values_tree_Fw



polished_actual_values_tree_Fw measured-versus-predicted TE collage

polished_actual_values_harmonic_regression_Fw

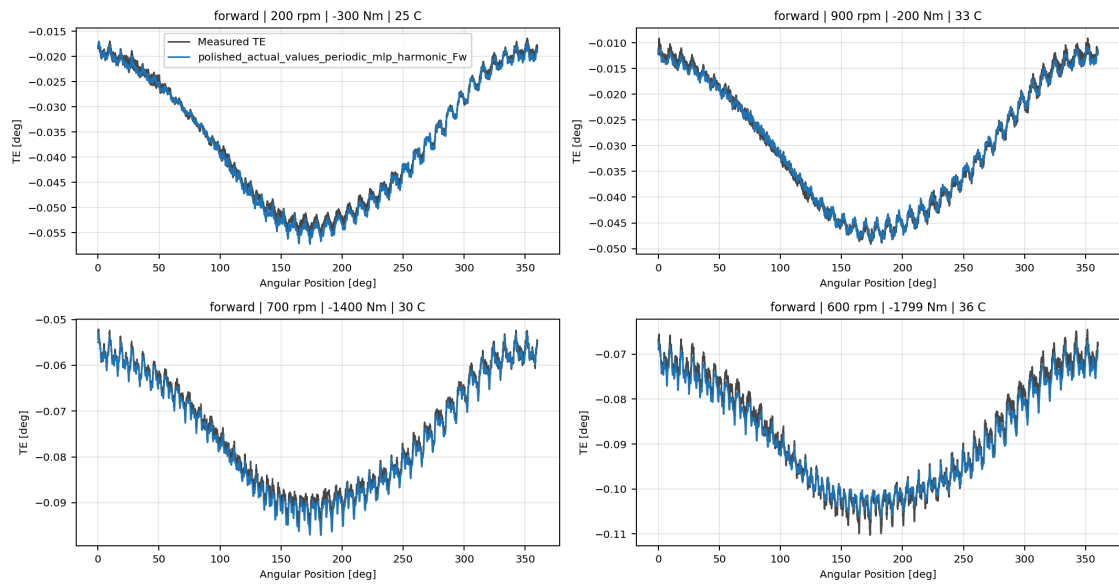
polished_actual_values_harmonic_regression_Fw



polished_actual_values_harmonic_regression_Fw measured-versus-predicted TE collage

polished_actual_values_periodic_mlp_harmonic_Fw

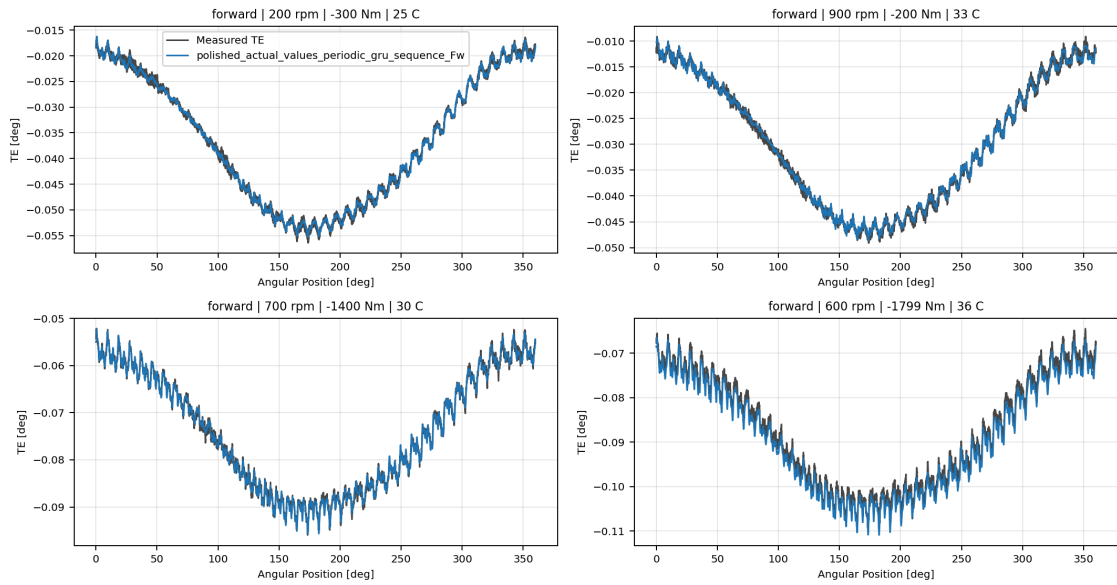
polished_actual_values_periodic_mlp_harmonic_Fw



polished_actual_values_periodic_mlp_harmonic_Fw measured-versus-predicted TE collage

polished_actual_values_periodic_gru_sequence_Fw

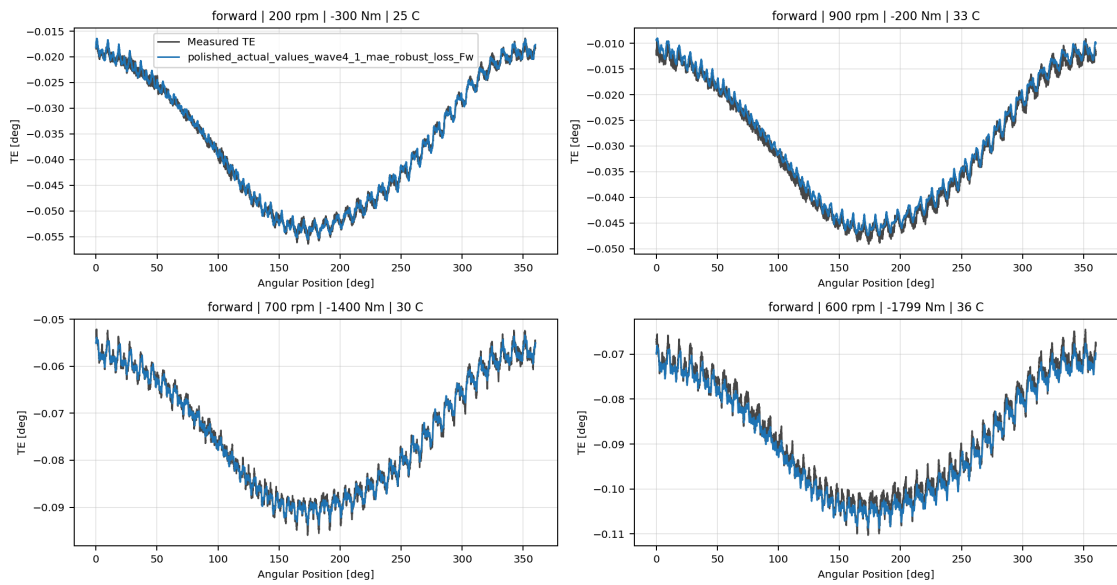
polished_actual_values_periodic_gru_sequence_Fw



polished_actual_values_periodic_gru_sequence_Fw measured-versus-predicted TE collage

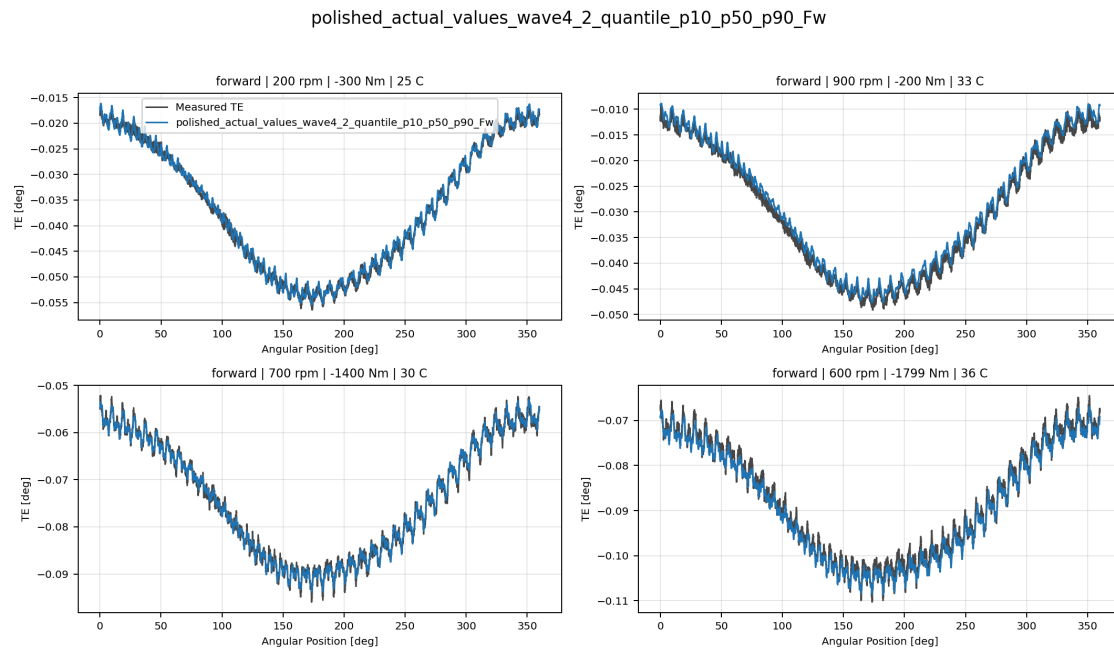
polished_actual_values_wave4_1_mae_robust_loss_Fw

polished_actual_values_wave4_1_mae_robust_loss_Fw



polished_actual_values_wave4_1_mae_robust_loss_Fw measured-versus-predicted TE collage

polished_actual_values_wave4_2_quantile_p10_p50_p90_Fw



polished_actual_values_wave4_2_quantile_p10_p50_p90_Fw measured-versus-predicted TE collage

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-19-17-46-52_track2_selected_active_polished_actual_values_matrix_track2_selected_active_polished_actual_values_forward_2026_07_19\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-19-17-46-52_track2_selected_active_polished_actual_values_matrix_track2_selected_active_polished_actual_values_forward_2026_07_19\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots\selected_active_model_matrix` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within this selected-active surface report. The curve-evidence section uses shared deterministic held-out operating conditions so visual shape fidelity can be compared across the selected families.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online `Table 9` compensation benchmark.
- RCIM Model-Bank Reproduction remains a separate paper-reference benchmark path and is not part of this selected-active model-only report.